

- Q.28 What is dynamic and static equalizing circuit? Why these are used in the circuit?
- Q.29 Give the block diagram of UPS and explain its basic parts.
- Q.30 Differentiate between Step up and Step down choppers.
- Q.31 What are the applications of Dual Converter?
- Q.32 What is an inverter? What are its different types?
- Q.33 Explain the working of single phase fully controlled full wave bridge rectifier.
- Q.34 Explain the working of SMPS with diagram.
- Q.35 Write merits and demerits of Series Inverter.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Define construction of SCR with diagram, explain working principle and modes of operation of SCR.
- Q.37 What is Chopper? Explain types of Choppers in detail.
- Q.38 What is a Cycloconverter? Explain single phase step up converter.

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Roll No.

5th Sem / IC, EI Sub : Power Electronics

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A SCR is a
- Two layer two junction device
 - Three layer two junction device
 - Four layer three junction device
 - Four layer four junction device
- Q.2 TRIAC is equivalent to
- Two SCRs connected in parallel
 - Two SCRs connected in anti-parallel
 - One diode and one SCR connected in parallel
 - One diode & one SCR connected in anti-parallel
- Q.3 AUJT is a
- Control current controlled device
 - Voltage controlled device option
 - Relaxation oscillator option
 - None of the above
- Q.4 A step down chopper can give an output voltage.
- Higher than input voltage
 - Lower than input voltage
 - Both higher and lower than input voltage
 - All of the above

- Q.5 Class A-Chopper operates in
 a) First quadrant b) Second quadrant
 c) All four quadrants d) Both (a) and (b)
- Q.6 MOSFET is a
 a) Current controlled device
 b) Both first and second quadrant
 c) Both (a) and (b)
 d) Power controlled device
- Q.7 An inverter converts
 a) DC into variable DC
 b) AC into DC
 c) DC into AC
 d) AC into AC different frequency
- Q.8 Cycloconverters drivers are generally used in
 a) Traction
 b) Milling
 c) Generating low frequencies
 d) Generation pulses
- Q.9 The holding current is _____ then the latching current
 a) Same b) Lower
 c) Higher d) None of the above
- Q.10 The output of regulated power supply
 a) Changes when into AC supply changes
 b) Remain constants a irrespective of Change in input AC change
 c) Changes when output load current changes
 d) Changes when both input as supply and output load current change

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The value of latching current is more than Holding Current. (True/False)
- Q.12 What is Triggering?
- Q.13 The three terminals of SCR are Gate. Drain and Source. (True/False)
- Q.14 Write any two applications of Dual Converters.
- Q.15 AUJT has _____ PN junction.
- Q.16 The duty cycle of a chopper is expressed as _____.
- Q.17 The full form of UPS is _____.
- Q.18 What is firing angle?
- Q.19 Armature Control method is used in DC motors to get _____.
- Q.20 Define Holding Current.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the V-I characteristics of a SCR.
- Q.22 Explain the controlling techniques of the chopper?
- Q.23 Explain the working of DIAC.
- Q.24 Explain different types of choppers according to quadrants.
- Q.25 Explain Centre Trapped Cycloconverter with the help of diagram.
- Q.26 Explain the UJT relaxation oscillator with diagram.
- Q.27 What are AC and DC drives?

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